BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 087 · Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is an invariant?

Q2. Why is sorting unnecessary for just finding one maximum?

Q3. What is nullable int?

Q4. Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

New learning

Observe — Specify output when inputs are empty, and preserve correctness on all-negative arrays.

Explain — Maintain invariant: best is the largest value among those inspected so far.

Evaluate — Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Worked example

Trace Maximum([7,3,10,4]).

1. Model response — The invariant tracks largest seen, not last examined.

We Do · Practise with guidance

Trace different Maximum([-9,-4,-11]).

Model response

Starting best at0 would be incorrect for all-negative arrays.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

Question 2

Write algorithm for maximum of array, including empty input case. [7]

Question 3

Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

Exit Ticket · Answer before checking

Initialising maximum to zero works for every integer array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A condition maintained while processing data.

Q2. A single scan can find it in O(n).

Q3. An int? that can be an integer or null.

Q4. mid2=6, then mid3=8; found index3.

You Do pages · Independent questions

Q1. Initial−5 [2], becomes−2 [2], remains−2 [1], returns−2 [1].

Q2. Empty return null [2], initial best=a[0] [1], iterate array [1], update if x>best [2], return best [1].

Q3. Scan O(n) [2], sorting typically O(n log n) [2], test empty and all-negative cases [2].

Exit Ticket · Model answer

False. It fails on arrays containing only negative values.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: